



# MADRIGAL ELECTROMOTIVES



# OUTLINE

- Business Problem
- Overview
- Data Understanding
- Data Analysis
- Graphs
- Recommendations
- Next Steps

# BUSINESS PROBLEM

- The Company is interested in expanding into the aviation industry, specifically in purchasing and operating airplanes for commercial and private enterprises.
- As the chief data scientist, it is my responsibility to identify which aircraft pose the lowest risk to the company in initiating this new business endeavor.
- My findings are illustrated in this presentation.



# OVERVIEW

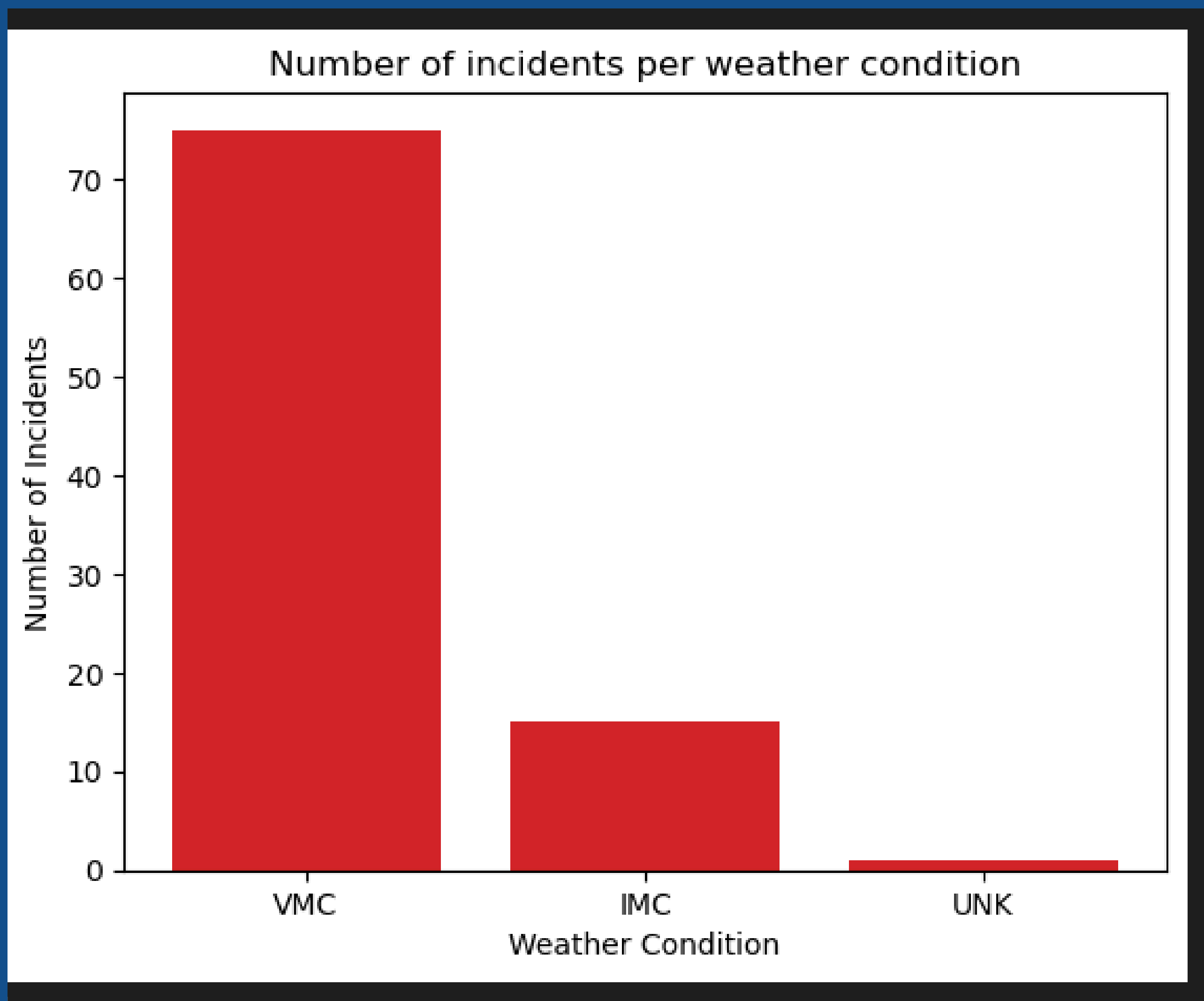
- I used data provided by the National Transportation Safety Board on aviation accidents to extract airplane accident data for thousands of airplanes in the market today.
- I identified columns of interest and removed the rest
- Using these columns of interest, I was able to find safe airplane registration numbers that I stored.
- I then identified the safest makes and models as well
- Lastly, I used weather and flight info to identify the best weather conditions to fly the airplanes in as well as the periods of flight that were the riskiest.

# DATA UNDERSTANDING

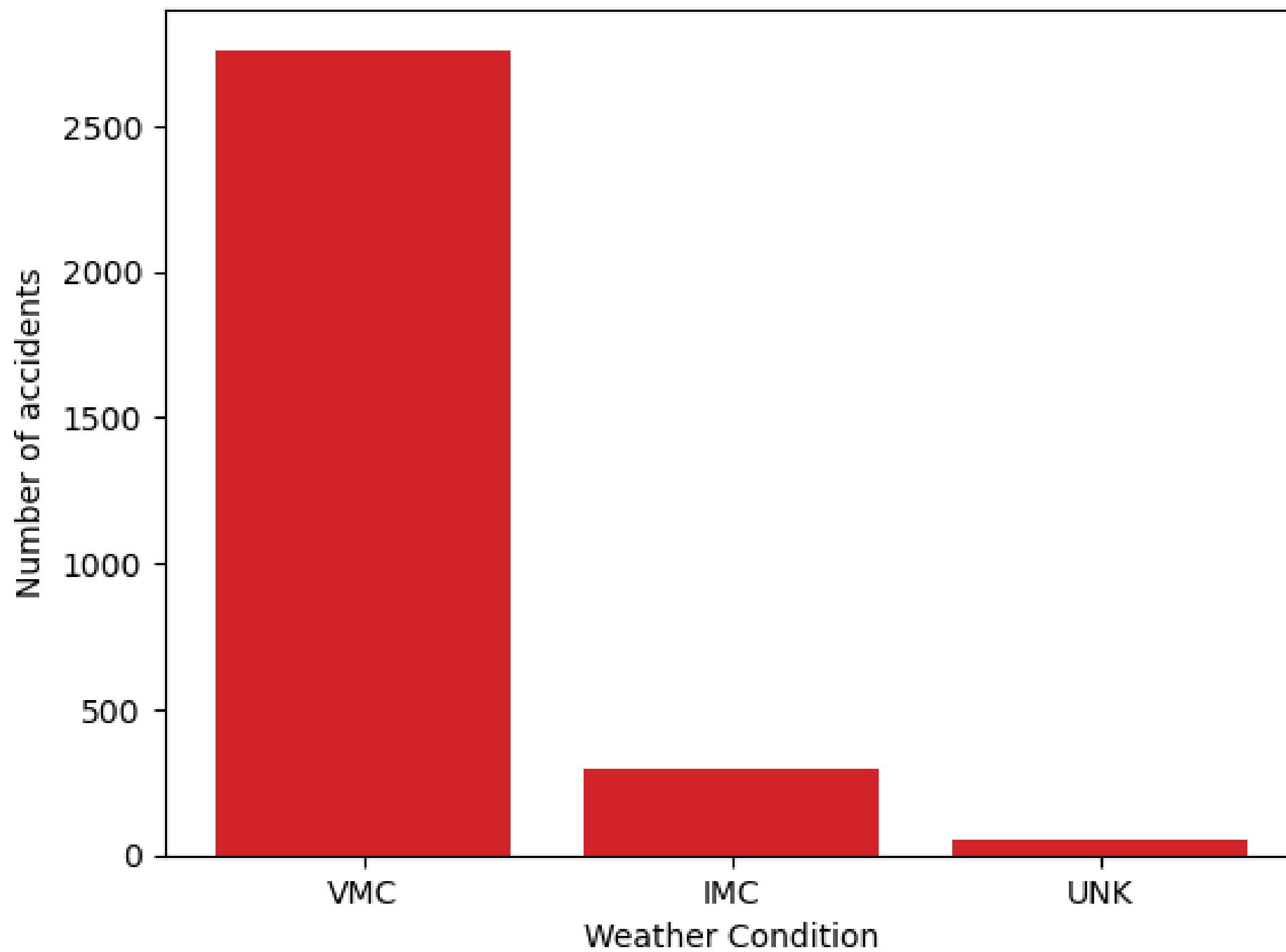
- The data provided was on aviation accidents that happened from 1962 to 2023
- It contained 90,348 records
- Every record had at least one missing value
- The dataset contained information on numerous aircraft. I selected only the airplane records.
- Some of the key columns I focused on were Registration Number, Total Fatal Injuries, and Total Uninjured
- Using these key columns I was able to drop any records with null column values and remained with 3201 records.
- It is these 3201 airplane records that I proceeded with.

# DATA ANALYSIS

- I isolated the airplanes that were involved in no accidents.
- They were all involved in only one incident.
- In the one incident they were involved in, I identified the number of injuries.
- I isolated the ones that had a low number of minor injuries and high uninjured.
- I classified these as super safe airplanes; airplanes with a low chance of injury were they to be involved in an accident/incident.
- I also identified the makes and models that appeared the least in the database and classified them as safe makes and models.

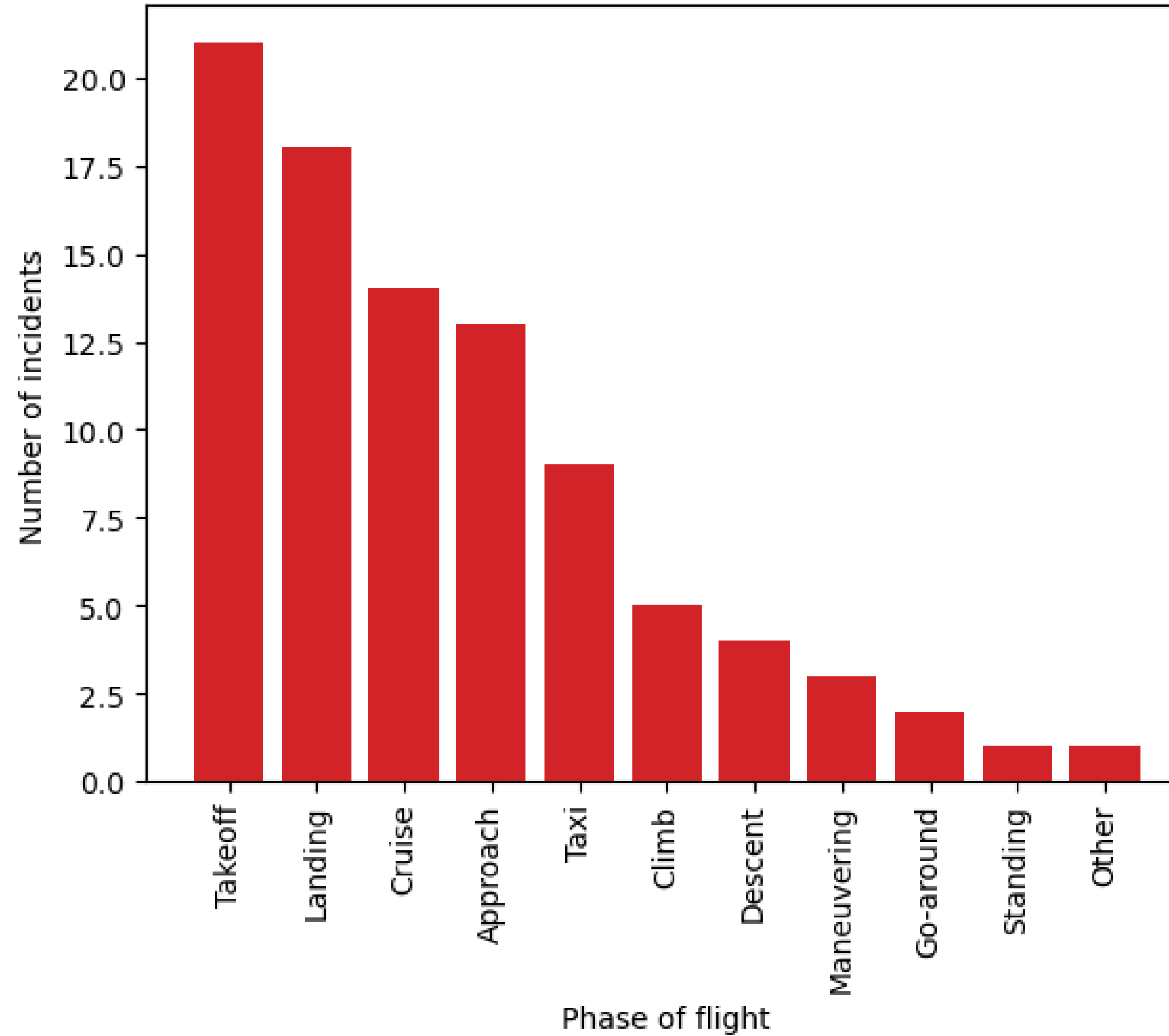


Number of accidents per weather condition

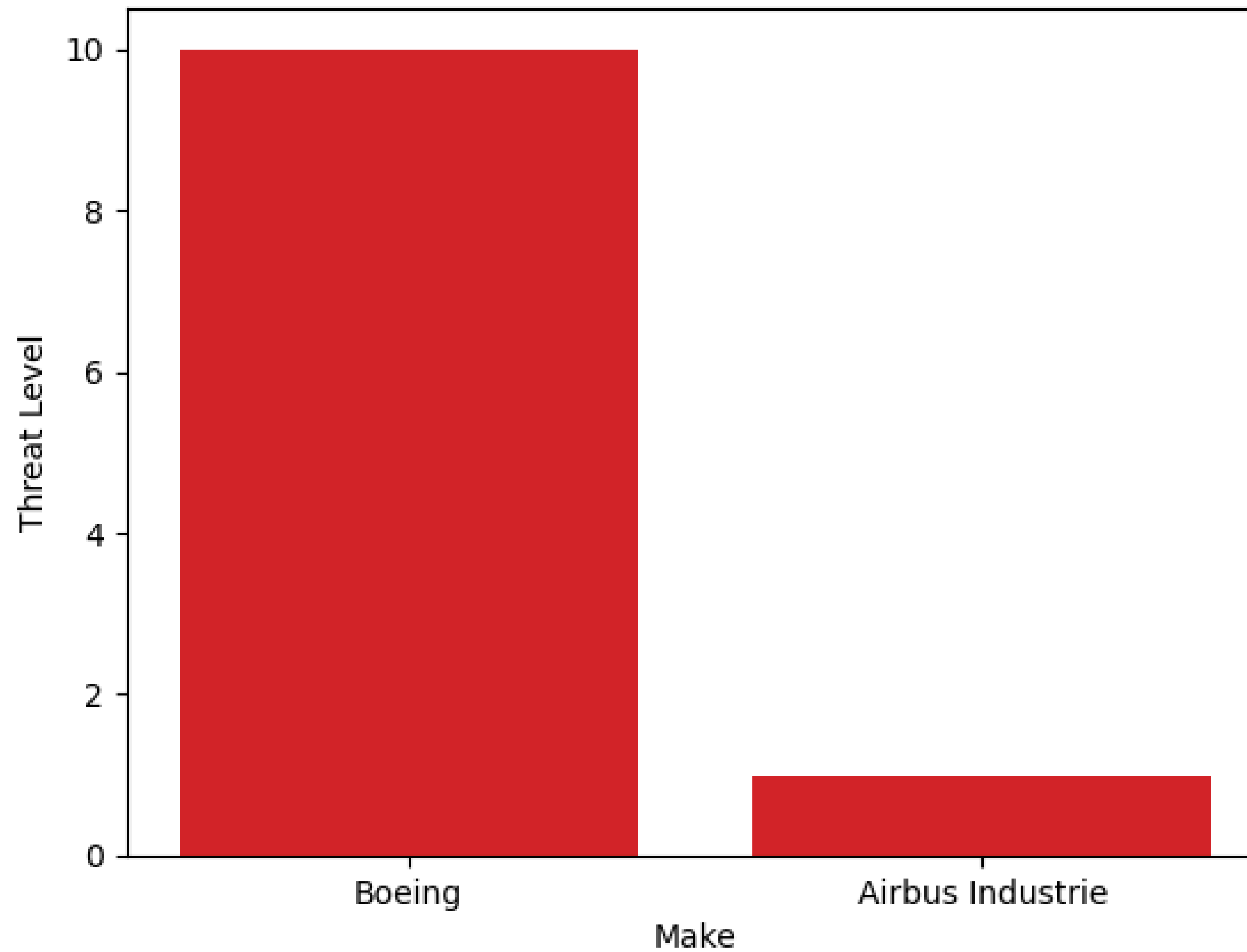




Number of incidents per phase of flight



Threat level per make of airplane



# RECOMMENDATIONS

- The company should only purchase the safe makes and models.
- Even better, if they are on sale, the airplanes with the registration numbers in the safe and super safe airplanes CSV.
- Pilots need to be very cautious when flying planes in Visual Meteorological Conditions weather.

# NEXT STEPS

- Follow up for a second opinion from pilots on what they think about the makes and models, as data only tells one side of the story.
- Liaise with airplane manufacturers for accident data on their planes in order to get better information.

THANK YOU

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